

Project Requirements

Functional Requirements

- Import retail sales dataset from Excel/CSV.
 - Clean and preprocess raw data using Python.
 - Perform Exploratory Data Analysis (EDA).
 - Generate business features using Feature Engineering.
 - Export cleaned dataset for SQL analysis.
 - Design a normalized MySQL database.
 - Load data using ETL (Staging Tables).
 - Perform business analysis using SQL.
 - Create reusable SQL Views and Stored Procedures.
 - Build an interactive Power BI dashboard.
 - Generate business insights and recommendations.
 - Maintain project documentation and GitHub repository.
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Non-Functional Requirements

- Data accuracy and consistency.
 - Optimized SQL queries.
 - Well-structured project folders.
 - Reusable SQL scripts.
 - Clear documentation.
 - Easy maintenance and scalability.
 - Professional code formatting.
 - Version control using Git.
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Hardware Requirements

- Processor: Intel Core i3 or above
 - RAM: Minimum 8 GB
 - Storage: Minimum 20 GB free space
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Software Requirements

Software	Version
Python	3.11+
Jupyter Notebook	Latest
MySQL Server	8.0+
MySQL Workbench	Latest
Power BI Desktop	Latest
Git	Latest
GitHub	Latest
Microsoft Excel	2019 / Microsoft 365

Python Libraries

- pandas
- numpy
- matplotlib
- openpyxl
- jupyter